

# Cassidy K. Buhler, Ph.D.

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in cassie-buhler

🐙 cassiebuhler.github.io/

🌐 cassiebuhler

## PROFESSIONAL APPOINTMENTS

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| 2026 – Present | <b>Postdoctoral Researcher</b><br><i>Eric and Wendy Schmidt Center for Data Science &amp; Environment (DSE)</i><br><i>University of California, Berkeley</i> | Berkeley, CA |
| 2024 – 2026    | <b>Postdoctoral Fellow</b><br><i>Environmental Data Science Innovation &amp; Impact Lab (ESILL)</i><br><i>University of Colorado, Boulder</i>                | Boulder, CO  |

## EDUCATION

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|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 2024 | <b>Ph.D. Operations Research</b><br>Computational Data Science Minor<br><i>Drexel University</i><br><i>Dissertation: Advances in Optimization with Applications to Biodiversity Conservation</i> | Philadelphia, PA   |
| 2019 | <b>B.S. Mathematics</b><br>Statistics Emphasis<br><i>University of Utah</i>                                                                                                                      | Salt Lake City, UT |

## PUBLICATIONS

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### Peer-Reviewed Journal Articles

**C. K. Buhler**, H. Y. Benson, and D. F. Shanno, “Regularized step directions in nonlinear conjugate gradient methods,” *Mathematical Programming Computation*, vol. 16, pp. 629–664, 2024, ISSN: 1867-2957. [DOI](#): 10.1007/s12532-024-00265-9.

**C. K. Buhler**, R. S. Terry, K. G. Link, and F. R. Adler, “Do mechanisms matter? Comparing cancer treatment strategies across mathematical models and outcome objectives,” *Mathematical Biosciences and Engineering*, vol. 18, no. 5, pp. 6305–6327, 2021, ISSN: 1551-0018. [DOI](#): 10.3934/mbe.2021315.

### Refereed Conference Proceedings

**C. K. Buhler** and H. Y. Benson, “Decision-making for land conservation: A derivative-free optimization framework with nonlinear inputs,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, 2024, pp. 21932–21939. [DOI](#): 10.1609/aaai.v38i20.30195.

**C. K. Buhler** and H. Y. Benson, “Optimal land conservation decisions for multiple species,” in *Proceedings of the 52nd Northeast Decision Sciences Institute Annual Conference*, vol. 52, Washington, D.C., 2023, pp. 808–816.

### Commentary & Perspectives

**C. K. Buhler**, F. Pérez, and C. Boettiger, “Why scientists should lead the shift away from AI mega data centres,” *Nature*, vol. 656, no. 8127, pp. 296–298, 2026. [DOI](#): 10.1038/d41586-026-02451-2.

### Policy & Technical Reports

C. Schloss, J. Howard, R. Olliff Yang, N. Roth, C. Dickison, A. Canales, C. K. Stanley, M. Webb, D. Ackerly, C. Boettiger, **C. Buhler**, S. Monfort, G. Bucciarelli, R. Meyer, A. Levi Simons, J. Bando, L. Chamberlin, S. Veloz, A. Hill, R. Johnson, R. Hill, A. Kochanek, D. Rademacher, L. Micheli, K. Penrod, G. Rapacciuolo, D. Spatz, and D. Gluesenkamp, “California’s 30x30 Initiative in 2025: A biodiversity assessment of the state’s 30x30 conservation areas,” Aug. 2026. [URL](https://cabiodiversitynetwork.org/wp-content/uploads/2026/08/CBN-Biodiversity-Assessment-2025-Final-Report.pdf): <https://cabiodiversitynetwork.org/wp-content/uploads/2026/08/CBN-Biodiversity-Assessment-2025-Final-Report.pdf>.

J. Howard, C. Schloss, R. Olliff Yang, A. Canales, C. K. Stanley, M. Webb, D. Ackerly, C. Boettiger, **C. Buhler**, S. Monfort, G. Bucciarelli, R. Meyer, A. Levi Simons, J. Bando, L. Chamberlin, S. Veloz, C. Dickison, A. Hill, R. Johnson, R. Hill, A. Kochanek, D. Rademacher, L. Micheli, K. Penrod, G. Rapacciuolo, N. Roth, D. Spatz, D. Gluesenkamp, and collaborators, "California's 30 x 30 initiative at its midpoint: A biodiversity assessment snapshot of the state's conservation areas," *Artemisia*, vol. 51, no. 2, Jan. 2026, California Native Plant Society.  URL: <https://www.cnps.org/artemisia-journal/californias-30-x-30-initiative-at-its-midpoint-44874>.

## SOFTWARE

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### **GLEN (Geospatial LLM-Enabled Navigator): A reusable library for map-based applications with LLM-powered data analysis**

 <https://github.com/boettiger-lab/geo-agent>

 10.5281/zenodo.20673849 (2026)

### **High Seas Explorer**

 <https://high-seas.nrp-nautilus.io>

 10.5281/zenodo.20674176 (2026)

### **Public GYE Explorer**

 <https://wyoming-public-demo.nrp-nautilus.io>

 10.5281/zenodo.20674178 (2026)

### **Trust for Public Land Explorer**

 <https://tpl.nrp-nautilus.io>

 10.5281/zenodo.18500622 (2026)

### **Trust for Public Land California Explorer**

 <https://tpl-ca.nrp-nautilus.io>

 10.5281/zenodo.20674186 (2026)

### **California 30x30 Planning & Assessment Tool (California Biodiversity Network Edition)**

 <https://huggingface.co/spaces/boettiger-lab/ca-30x30-cbn>

 10.5281/zenodo.16988638 (2025)

### **LandVote LLM Decision-Support Prototype**

 <https://huggingface.co/spaces/boettiger-lab/landvote>

 10.5281/zenodo.18500795 (2026)

### **California 30x30 Planning & Assessment Prototype**

 <https://huggingface.co/spaces/boettiger-lab/ca-30x30>

 10.5281/zenodo.14933818 (2025)

### **Derivative-Free Optimization for Land Conservation**

 <https://github.com/cassiebuhler/conservation-dfo>

 10.5281/zenodo.13742960 (2024)

### **Conmin-CG: Hybrid Cubic Regularization of Conjugate Gradient Methods**

 <https://github.com/cassiebuhler/ConminCG>

 10.5281/zenodo.13315592 (2024)

## DATASETS

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### **State-level 30x30 conservation policy implementation and progress across the United States**

 <https://source.coop/cassiebuhler/30x30-state-policy>

 10.5281/zenodo.18500907 (2026)

## WORKING GROUPS

- 2026 – **EnviroLLM: Developing Best Practices for Large Language Models in Environmental Science**  
Present  <https://cu-esiil.github.io/envirollm/>
- 2024 – **California Biodiversity Network**  
Present  <https://cabiodiversitynetwork.org/>
- 2025 **Maka Sitomniya: Preserving Mother Earth by Asserting Lakota Sovereignty in Earth Data Science**  
 <https://cu-esiil.github.io/Maka-Sitomniya/>

## FELLOWSHIPS & RESEARCH EXPERIENCE

- 2024 – **Postdoctoral Fellowship (NSF Award Number: 2153040)** Boulder, CO  
2026 *Environmental Data Science Innovation & Impact Lab (ESIIL)*  
*University of Colorado, Boulder*
- 2019 – **Doctoral Research Fellow** Philadelphia, PA  
2024 *Decision Sciences & MIS Department*  
*Drexel University*
- 2019 – **Research Assistant** Salt Lake City, UT  
2021 *Adler Lab - Mathematics Department*  
*University of Utah*
- 2018 – **Undergraduate Research Assistant** Salt Lake City, UT  
2019 *Research Experience for Undergraduates (REU)*  
*University of Utah*
- 2018 **Computer Scientist (Internship)** Hill AFB, UT  
*309th Software Engineering Group*  
*United States Air Force*

## TEACHING

- 2019 – **Instructor** Philadelphia, PA  
2024 *Decision Sciences & MIS Department*  
*Drexel University*

| Course                                                       | Level | Quarter(s)                        | Tool(s)                |
|--------------------------------------------------------------|-------|-----------------------------------|------------------------|
| BSAN 360: Programming for Data Analytics                     | U     | Winter 2022                       | R                      |
| Ph.D. Programming Bootcamp                                   | PhD   | Summer 2021; Summer 2022          | Python                 |
| MIS 200: Management Information Systems (Recitation Section) | U     | Fall 2019; Fall 2020; Winter 2021 | MS Access; Excel; HTML |

\*Undergraduate (U)

## TEACHING (CONTINUED)

2019 – **Teaching Assistant** Philadelphia, PA  
 2024 *Decision Sciences & MIS Department*  
*Drexel University*

| Course                                                      | Level     | Quarter(s)                          | Tool  |
|-------------------------------------------------------------|-----------|-------------------------------------|-------|
| BSAN 360: Programming for Data Analytics                    | U         | Spring 2021                         | R     |
| BSAN 601: Business Analytics for Managers                   | MS; MBA   | Spring 2024                         | Excel |
| MIS 612: Aligning Information Systems & Business Strategies | EMBA; MBA | Fall 2023                           | -     |
| MIS 625: Management of IT Operations                        | MBA       | Fall 2023                           | -     |
| OPM 200: Operations Management                              | U         | Spring 2020; Fall 2021; Spring 2023 | Excel |
| OPM 341: Supply Chain Management                            | U         | Spring 2021; Spring 2022; Fall 2022 | Excel |
| OPM 344: Revenue Management                                 | U         | Fall 2022                           | Excel |
| OPR 320: Linear Models for Decision Making                  | U         | Summer 2020; Spring 2021            | Excel |

### Teaching Assistant (Continued)

| Course                                               | Level   | Quarter(s)                                                                 | Tool  |
|------------------------------------------------------|---------|----------------------------------------------------------------------------|-------|
| STAT 201: Intro to Business Statistics               | U       | Winter 2020; Spring 2020; Fall 2021; Summer 2022; Spring 2023; Winter 2024 | Excel |
| STAT 202: Business Statistics II                     | U       | Summer 2021; Spring 2023                                                   | Excel |
| STAT 205: Statistical Inference I                    | U       | Spring 2020; Fall 2021                                                     | Excel |
| STAT 206: Statistical Inference II                   | U       | Summer 2021                                                                | Excel |
| STAT 510: Intro to Statistics for Business Analytics | MBA     | Summer 2023; Winter 2024                                                   | Excel |
| STAT 642: Data Mining for Business Analytics         | MS; PhD | Winter 2023                                                                | R     |

\*Undergraduate (U)

2018 – **Mathematics & Computer Lab Assistant** Salt Lake City, UT  
 2019 *T. Benny Rushing Mathematics Student Center*  
*University of Utah*

## PRESENTATIONS

2026 **California 30x30 Partnership 2026 Summit** San Francisco, CA  
 Workshop: From Biodiversity Science to Conservation Action

2026 **Deep Ocean Collective Solution Accelerator Workshop 2026** La Jolla, CA  
 Talk: Using AI to move from ocean observations to decision-ready information

2026 **ESA Annual Meeting 2026** Salt Lake City, UT  
 Talk: Collaborative conservation planning with AI decision-support tools

2026 **Mountain Research Station (MRS) Summer Seminar Series** Nederland, CO  
 Talk: Responsibly using AI for conservation decision-making

2026 **Natural Capital Symposium 2026** Stanford, CA  
 Talk: Codeveloping AI decision-support tools to democratize conservation planning

2025 **California 30x30 Partnership 2025 Summit** San Diego, CA  
 Talk: Metrics with Meaning: Assessing, Tracking, and Supporting 30x30 Biodiversity Conservation

## PRESENTATIONS (CONTINUED)

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|      |                                                                                                                                                                                 |                         |
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| 2025 | <b>NASA Biodiversity and Ecological Conservation Team Meeting</b><br>Poster: Leveraging NASA Data to Guide Biodiversity Conservation Investments with the Trust for Public Land | Washington, DC.         |
| 2024 | <b>AGU Annual Meeting (AGU24)</b><br>Poster: Exploring innovation in biodiversity conservation decision-making through open science and generative AI                           | Washington, DC.         |
| 2024 | <b>AAAI Conference on Artificial Intelligence (AAAI-24)</b><br>Poster: Decision-making for land conservation: A derivative-free optimization framework with nonlinear inputs    | Vancouver, BC, Canada.  |
| 2023 | <b>MIT Sloan Rising Scholars Conference</b><br>Talk: Decision-making for land conservation: A derivative-free optimization framework with nonlinear inputs                      | Cambridge, MA (Virtual) |
| 2023 | <b>INFORMS Annual Meeting</b><br>Talk: Decision-making for land conservation: A derivative-free optimization framework with nonlinear inputs                                    | Phoenix, AZ.            |
| 2023 | <b>SIAM Conference on Optimization (OP23)</b><br>Talk: Reserve design in biodiversity conservation                                                                              | Seattle, WA.            |
| 2023 | <b>NEDSI Annual Conference</b><br>Talk: Optimal land conservation decisions for multiple species                                                                                | Washington, D.C.        |
| 2021 | <b>INFORMS Annual Meeting</b><br>Talk: Regularized step directions in conjugate gradient minimization for machine learning                                                      | Anaheim, CA. (Virtual)  |
| 2021 | <b>SIAM Conference on Optimization (OP21)</b><br>Talk: Conjugate gradient methods for machine learning                                                                          | Virtual.                |
| 2020 | <b>INFORMS Annual Meeting</b><br>Talk: Efficient solution of portfolio optimization problems via dimension reduction and sparsification                                         | Virtual.                |

## MEDIA

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|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| 2026 | <b>The Mountain-Ear</b><br>Podcast Interview: Inside CU's world-class alpine research center<br> <a href="https://podcasts.apple.com">podcasts.apple.com</a> | Nederland, CO |
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## AWARDS & GRANTS

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|------|----------------------------------------------------------------------------------------------------|
| 2023 | <b>Rising Scholar</b><br><i>MIT Sloan School of Management</i>                                     |
| 2023 | <b>Graduate Student Travel Subsidy Award</b><br><i>Drexel University</i>                           |
| 2023 | <b>DEI &amp; Environment and Sustainability Innovation Micro-Grant</b><br><i>Drexel University</i> |

## AWARDS & GRANTS (CONTINUED)

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|------|---------------------------------------------------------------------------------------------|
| 2023 | <b>Teck-Kah Lim Graduate Student Travel Subsidy Award</b><br><i>Drexel University</i>       |
| 2023 | <b>Student Travel Award</b><br><i>Society for Industrial and Applied Mathematics (SIAM)</i> |
| 2022 | <b>Teaching Assistant Excellence Award</b><br><i>Drexel University</i>                      |
| 2021 | <b>Teaching Assistant Excellence Award (Highly Commended)</b><br><i>Drexel University</i>   |
| 2021 | <b>Student Travel Award</b><br><i>Society for Industrial and Applied Mathematics (SIAM)</i> |
| 2019 | <b>Undergraduate Research Scholar Designation</b><br><i>University of Utah</i>              |

## SERVICE

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|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| 2025             | <b>Panelist</b><br>Event: Femme in STEM                                                                                                                                                                | <i>CU Boulder, Career Services</i>                                 |
| 2024-<br>Present | <b>CIRES Science Pathways Speaker</b><br>• Promoting science engagement by delivering flash talks at Colorado institutions                                                                             | <i>Cooperative Institute for Research in Environmental Science</i> |
| 2023             | <b>Session Chair</b><br>Session: Nonlinear Optimization in Machine Learning                                                                                                                            | <i>INFORMS Annual Meeting</i>                                      |
| 2023             | <b>Session Organizer</b><br>Session: Nonlinear Optimization and Applications                                                                                                                           | <i>SIAM Conference on Optimization</i>                             |
| 2023             | <b>Session Chair</b><br>Session: Land, Sand, and Plastic Management                                                                                                                                    | <i>NEDSI Annual Conference</i>                                     |
| 2022             | <b>Panelist</b><br>Event: Graduate Teaching Assistance Orientation                                                                                                                                     | <i>Drexel University</i>                                           |
| 2019             | <b>Mathematics Tutor (Volunteer)</b><br>• Provided weekly tutoring sessions at the Utah State Prison<br>• Supported students who are incarcerated and taking a Salt Lake Community College math course | <i>Utah Prison Education Project</i>                               |